

SPEECH UNDERSTANDING AS A TRAINING SIGNAL FOR GENERATION: SHARED SUPERVISION AND LATENT RECONSTRUCTION FEEDBACK

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ABSTRACT

Speech understanding learns the correspondence between speech and text, motivating its use as an auxiliary task for speech generation. However, transcription supervision on real audio does not directly assess generated content. We combine shared text prediction with Reconstruct CE, which differentially reconstructs speech latents from the flow-training state and applies transcript supervision through a frozen understanding branch. This connects local velocity learning with supervision of reconstructed content, without waveform decoding in the reconstruction branch or changes to inference. At both 10:1 and 20:1 generation-to-understanding ratios, matched continuations reduce Chinese CER by 10.7–12.7% and English WER by 9.5–10.4% relative to generation-only continuation. A separate matched-update ablation shows that Reconstruct CE reduces these errors by a further 6.4% and 8.0% relative to shared supervision, while improving predicted naturalness and speaker similarity. The converged model achieves higher speaker similarity than six re-evaluated baselines in both languages under their respective generation settings. Our approach thus uses speech understanding to supervise speech generation, transferring linguistic knowledge through shared training and providing direct feedback on reconstructed content. Audio samples are available online.

Index Terms— speech generation, understanding supervision, flow matching, text reconstruction, multi-task learning

1. INTRODUCTION

Text-to-speech (TTS) synthesis aims to express a given text as natural speech, with reference-conditioned systems also preserving the speaker’s voice. Flow matching [1] supports non-autoregressive synthesis in F5-TTS [2] and broader audio generation in MMAudio [3]. These models predict local velocities that progressively transform noise into audio. Alongside acoustic fidelity, accurate realization of the intended text remains essential, motivating complementary training signals that explicitly supervise the content of generated speech.

Speech understanding offers a source of such supervision by learning the reverse audio-to-text correspondence. Unified models provide a way to transfer this knowledge through shared parameters: SpeechT5 [4] uses encoder–decoder pretraining, VioLA [5] unifies

recognition, synthesis, and translation through autoregressive codec language modeling, and UniVoice [6] combines autoregressive recognition with flow-based synthesis. UniAudio 2.0 and UALM extend joint learning across audio understanding and generation tasks [7, 8]. These developments motivate retaining an understanding objective when training primarily for generation. However, a transcription loss evaluated only on real audio does not directly assess generated content. This leaves open how knowledge learned by the understanding branch can also supervise the generator’s own predictions.

Prior work suggests that analyzing synthesized audio can provide additional feedback. The machine speech chain [9] reconstructs text from synthesized speech; StyleTTS 2 [10] uses pretrained speech-model representations for adversarial supervision; and Cont-SPT [11] applies ASR supervision to reconstructed audio during tokenizer pretraining. Applying transcript feedback during flow-based training requires connecting two levels of supervision: flow matching evaluates a local velocity at a sampled state, whereas transcription evaluates content at the audio endpoint. A differentiable reconstruction trajectory can connect endpoint feedback to the velocity predictions used in flow training. We thus ask: **How can learning to understand speech improve learning to generate speech?** Specifically, we examine transfer through both a shared training objective and direct feedback on reconstructed content.

We investigate this question through shared text prediction and Reconstruct CE. Shared text prediction retains understanding supervision during generation-focused training. Reconstruct CE applies transcript supervision to differentially reconstructed speech latents through a frozen understanding branch, providing content feedback to the generator. This branch operates entirely in latent space and is used only during training. We evaluate both paths in a unified MMDiT-based speech understanding and generation model using matched bilingual continuations. These experiments separately assess the benefit of retaining understanding supervision and the additional benefit of reconstruction feedback. Both paths improve content accuracy, with frozen-reader feedback also improving predicted naturalness and speaker similarity over shared supervision.

2. UNDERSTANDING SUPERVISION FOR GENERATION

Figure 1 summarizes the two training paths: text prediction on real audio and transcript feedback on reconstructed latents. The latter propagates gradients through reconstruction; neither path changes generation-time decoding.

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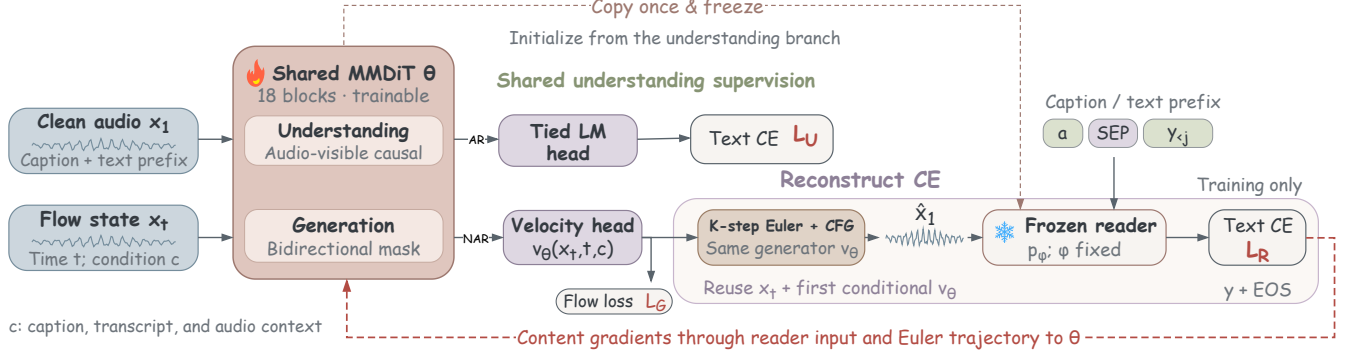


Fig. 1. Understanding supervision for speech generation. Shared learning combines flow matching with text prediction on real audio. Reconstruct CE reuses the flow-training state and a first-step conditional velocity prediction for K differentiable CFG-guided Euler steps. Transcript gradients reach the generator through the frozen reader’s audio input and the reconstruction trajectory. Reconstruction runs only during training. a/y : caption/transcript; AR/NAR: autoregressive/non-autoregressive. Waveforms represent audio latents.

2.1. Problem Formulation and Backbone

Let x_1 , y , and a denote clean audio latents, transcript, and caption. Generation conditions on $c = [a, \text{SEP}, y]$ and available audio context. Our MMDiT has 1.69B parameters, 18 joint blocks, a hidden width of 1024, and 64-dimensional audio latents.

The two tasks share the backbone, embeddings, and projections, including a tied text output head. Following UniVoice [6], generation uses bidirectional audio–text attention. For understanding, audio tokens attend only to audio, while text tokens attend to audio and preceding text, preventing future-text leakage.

2.2. Shared Understanding Supervision

For Gaussian noise ϵ and $t \sim \mathcal{U}(0, 1)$, flow matching uses $x_t = tx_1 + (1 - t)\epsilon$. The generation loss $\mathcal{L}_G$ is the mean squared error between $v_\theta(x_t, t, c)$ and $x_1 - \epsilon$, averaged over valid predicted frames and latent dimensions. Known audio-context frames remain fixed to their clean latents and are excluded from the prediction loss.

The same backbone learns to predict text from clean audio at $t = 1$. The understanding loss $\mathcal{L}_U$ is teacher-forced cross-entropy, with caption, separator, transcript, and EOS tokens as targets. We combine the objectives as $\mathcal{L}_{\text{joint}}^{(r)} = (r\mathcal{L}_G + \mathcal{L}_U)/(r + 1)$. Here r is the generation-to-understanding loss-weight ratio. Asymmetric weights prioritize generation while retaining auxiliary linguistic supervision. Under this normalization, moving from 10:1 to 20:1 mainly reduces the understanding coefficient and only modestly increases the generation coefficient. Reconstruction experiments start from a separate joint checkpoint, as specified in Sec. 3.1.

2.3. Differentiable Latent Reconstruction

Reconstruct CE integrates forward from the sampled flow-training state to obtain a differentiable audio endpoint. Each update combines conditional and unconditional velocity predictions using classifier-free guidance (CFG):

$$v_\theta^{\text{cfg}}(z, t, c) = (1 + g_{\text{rec}})v_\theta(z, t, c) - g_{\text{rec}}v_\theta(M_\emptyset z, t, c_\emptyset), \quad (1)$$

where g_{rec} is the reconstruction guidance strength. The masking operator $M_\emptyset$ zeros only known audio-context positions in z , preserving

the region to be generated. For $c_\emptyset$, all text embeddings, including caption and transcript, and any separate reference-audio latents are zeroed; lengths and masks remain unchanged.

With $z_0 = x_t$ and $h = (1 - t)/K$, we obtain $\hat{x}_1 = z_K$ through K differentiable Euler steps:

$$z_{k+1} = \Pi_c \left[z_k + h v_\theta^{\text{cfg}}(z_k, t + kh, c) \right], \quad (2)$$

for $k = 0, \dots, K - 1$. Here, Π_c restores clean known audio context and clears padding at every step before the next update.

At the first Euler step, we reuse the conditional velocity prediction used in the flow loss and compute an additional unconditional prediction. All subsequent steps use the same CFG rule. Both velocity branches remain differentiable throughout the trajectory.

2.4. Frozen-Reader Content Supervision

We initialize reader p_ϕ once from the same joint checkpoint as the generator and keep its parameters frozen. Transcript-loss gradients pass through the frozen reader’s audio input and the reconstruction trajectory to update the generator. At $t = 1$, the reader receives $\hat{x}_1$, which combines fixed audio context with the reconstructed region. Its target y is the transcript of the complete utterance, not only the generated span:

$$\mathcal{L}_R = -\frac{1}{N_y} \sum_{j \in \mathcal{T}_y} \log p_\phi(y_j | \hat{x}_1, a, y_{<j}), \quad (3)$$

where $\mathcal{T}_y$ contains transcript and EOS positions and N_y counts valid target tokens. Caption and separator tokens provide context. Samples with dropped text conditioning still contribute to flow matching but are excluded from $\mathcal{L}_R$, so transcript feedback is applied only to text-conditioned reconstruction.

Reconstruction training optimizes

$$\mathcal{L}_{\text{recon}} = \frac{10}{12}\mathcal{L}_G + \frac{1}{12}\mathcal{L}_U + \frac{1}{12}\mathcal{L}_R. \quad (4)$$

CE-off removes only $\mathcal{L}_R/12$, keeping the other coefficients fixed; the unfrozen-reader ablation also updates ϕ . The reconstruction branch operates entirely on latents and is removed at inference.

Table 1. Seed-TTS-eval results. A: re-evaluated baselines and our converged model; B: matched continuations from 300k for 200k updates. All systems use the same evaluation set and metric implementations, with CER for Chinese and WER for English. Acoustic metrics are reported when available. MOS averages 160 ratings per system/language (20 listeners $\times$ 8 utterances; Sec. 3.3). Bold marks the best values within each panel, including ties.

Model / setting	Params (B)	Chinese				English			
		CER (%) ↓	SIM ↑	UTMOS ↑	MOS ↑	WER (%) ↓	SIM ↑	UTMOS ↑	MOS ↑
<i>A. Re-evaluated baselines and our reconstruction model</i>									
Qwen2.5-Omni-7B	10.73	2.74	0.75	3.53	4.50	5.60	0.63	4.19	4.06
MiMo-Audio-7B-Instruct	8.02	3.10	0.58	2.66	3.06	4.03	0.44	3.54	3.13
Step-Audio 2-mini	8.32	1.41	0.64	3.51	3.88	1.89	0.47	4.31	3.75
UniAudio 2.0	4.86	5.63	0.48	3.09	3.00	4.76	0.34	3.84	3.31
Ming-Lite-Omni	0.69 (Talker only)	2.05	0.61	3.73	3.19	5.62	0.45	4.24	3.75
Ming-UniAudio-16B-A3B	18.27	1.06	0.70	2.77	3.56	1.82	0.58	3.67	3.44
Reconstruct CE (converged)	1.69	1.69	0.77	3.35	3.75	3.02	0.68	3.94	3.88
<i>B. Controlled continuations with shared supervision</i>									
Generation-only continuation	1.69	2.52	0.73	3.12	—	4.23	0.67	3.76	—
Joint 10:1	1.69	2.20	0.74	3.18	—	3.83	0.67	3.80	—
Joint 20:1	1.69	2.25	0.74	3.20	—	3.79	0.68	3.80	—

3. EXPERIMENTAL SETUP

3.1. Data and Controlled Training

The training mixture contains 26.23M entries, including replay, with a 60/40 Chinese/English split. We use caption-plus-transcript conditioning, random-span inpainting, and text-conditioning dropout of 0.2. Seed-TTS-eval [12] contains 2020 Chinese and 1088 English cases. Controlled comparisons use the same reference audio, transcripts, and assignment of two fixed captions per language.

Two continuation studies separate the contributions of shared supervision and reconstruction feedback. In Table 1B, all branches start from Joint 10:1 300k and train for another 200k updates: generation-only removes understanding supervision, Joint 10:1 retains it, and Joint 20:1 changes the generation-to-understanding loss-weight ratio from 10:1 to 20:1. In Table 2, all variants start from Joint 10:1 800k EMA and train for 100k updates. CE-off retains the flow and understanding coefficients; the two reader variants add reconstruction with identical $K = 10$, $g_{\text{rec}} = 4$, and loss weights. The converged frozen-reader model in Table 1A uses the same settings for 300k reconstruction updates (1.1M total).

3.2. Baseline Evaluation

We re-evaluate six released systems: Qwen2.5-Omni-7B [13], MiMo-Audio-7B-Instruct [14], Step-Audio-2-mini [15], UniAudio 2.0 [7], Ming-Lite-Omni [16], and Ming-UniAudio-16B-A3B [17]. Qwen uses the fixed Chelsie voice without reference conditioning. Its Thinker uses transcript-constrained greedy decoding, and its Talker has a 4096-token limit. MiMo uses reference-prompt TTS with a voice-matching instruction (global/local temperatures 0.6/0.9; top- p 1.0/0.95). Step uses temperature 0.7, a 4096-token limit, and reference-conditioned token-to-wave decoding.

Ming-Lite’s zero-shot Talker uses reference speech, transcript, and speaker features. UniAudio pairs `llm_ep2` with Reasoning-Codec; `prompt_instruct_tts` receives reference tokens, target text, and “Natural speech with a neutral speaking style.” (temperature 0.9, top- k 50, no CFG, 10 codec steps). Ming-UniAudio uses native text normalization and reference audio/transcript (10 diffusion iterations, CFG=2, 16-kHz output).

Each baseline produces one output per case using an utterance-specific seed derived from 42. Ming-Lite yields 1086 English outputs after two decoder failures; all other evaluations are complete. One UniAudio case with invalid codec output is retried with another seed and retained. All systems use the same evaluators, although their generation interfaces differ. Qwen’s fixed-voice SIM is not a matched voice-cloning comparison.

Parameter counts exclude standalone codecs/vocoders and our training reader, but include Qwen’s integrated token-to-wave module. For Ming-Lite, we count only the Talker; for MoE models, we count all experts. Tied weights are counted once.

3.3. Generation and Metrics

For each system and language, 20 listeners each rate eight utterances in a blinded evaluation on a 1–5 quality scale. MOS is the arithmetic mean of the resulting 160 ratings.

Our model uses EMA weights and per-utterance sampling with $v_{\text{cfg}} = (1 + g)v_{\text{cond}} - gv_{\text{uncond}}$, $g = 4$, and zeroed reference in the unconditional branch. We use 48 Euler steps for Chinese and 64 for English, time shift 1, and the same VAE posterior-sampling procedure for reference latents.

We compute Chinese CER with Paraformer [18] and English WER with Whisper large-v3 [19] through *faster-whisper*. SIM uses WavLM-Large features [20] with ECAPA-TDNN [21] (`wavlm_large_finetune.pth`) and cosine similarity to paired reference audio; UTMOS [22] predicts naturalness. Automatic metrics are averaged over utterances. Scores for our checkpoints in Tables 1–2 are further averaged over three inference seeds.

3.4. Auxiliary Protocols

The historical trajectories in Fig. 2 use learning rate 5×10^{-4} , seed 42, dynamic batches, and 2048 understanding tokens. Generation-only and Joint 10:1 share initialization; Joint 20:1 restarts from Joint 10:1 at 300k. The sampling-depth pilot in Fig. 3 uses CFG=4, batches of five, preserved punctuation, and historical reference audio.

Table 2. Reconstruction ablation: 100k updates from Joint 10:1 800k EMA, averaged over three inference seeds. Both reader variants use $K = 10$ and reconstruction guidance $g_{\text{rec}} = 4$. CE-off removes only $\mathcal{L}_R/12$ while preserving the other loss coefficients.

Training setting	Chinese			English		
	CER (%) ↓	SIM ↑	UTMOS ↑	WER (%) ↓	SIM ↑	UTMOS ↑
CE-off (shared supervision)	1.88	0.75	3.18	3.39	0.65	3.86
Reconstruct CE (unfrozen reader)	1.86	0.77	3.16	3.52	0.67	3.85
Reconstruct CE (frozen reader)	1.76	0.77	3.27	3.12	0.67	3.90

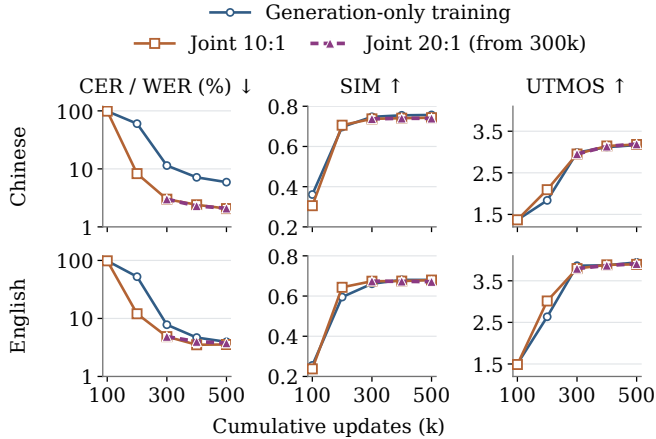


Fig. 2. Historical trajectories (seed 42; log CER/WER). Joint 20:1 branches at 300k with optimizer/data progress reset.

4. RESULTS AND DISCUSSION

Shared understanding supervision improves generation. Table 1B compares three continuations from Joint 10:1 300k, each trained for another 200k updates. Joint 10:1 reduces Chinese CER/English WER from 2.52%/4.23% to 2.20%/3.83%. Joint 20:1 reaches 2.25%/3.79% (10.7%/10.4% relative reductions), with UTMOS rising from 3.12/3.76 to 3.20/3.80. Both joint-learning settings improve content accuracy over generation-only continuation. Smaller, language-dependent differences between ratios suggest that the benefit persists across both tested weightings.

Reconstruction adds complementary content feedback. The frozen-reader variant reduces CER/WER from 1.88%/3.39% to 1.76%/3.12% versus CE-off (6.4%/8.0% relative; Table 2). It also raises UTMOS from 3.18/3.86 to 3.27/3.90 and SIM from 0.75/0.65 to 0.77/0.67. These gains are consistent with endpoint transcript feedback complementing local velocity supervision.

Compared with the frozen reader, the trainable reader increases CER/WER to 1.86%/3.52%, with unchanged SIM at the reported precision. Relative to CE-off, it slightly improves Chinese CER but worsens English WER and lowers UTMOS in both languages. These results favor a fixed reader in this setting. Freezing keeps the audio-to-text mapping fixed, directing reconstruction feedback to the generator. Independent waveform recognizers assess content accuracy, so the gains extend beyond agreement with the training reader.

Converged accuracy and speaker preservation. The converged model reaches 1.69% CER and 3.02% WER, with the highest SIM in both languages in Table 1A (0.77 and 0.68, respectively). It improves all three automatic metrics over MiMo-Audio and UniAudio 2.0 in both languages. Relative to Ming-UniAudio, it has higher CER/WER but higher SIM (0.77/0.68 versus 0.70/0.58) and UTMOS (3.35/3.94 versus 2.77/3.67). Its MOS is not the highest, distinguishing speaker

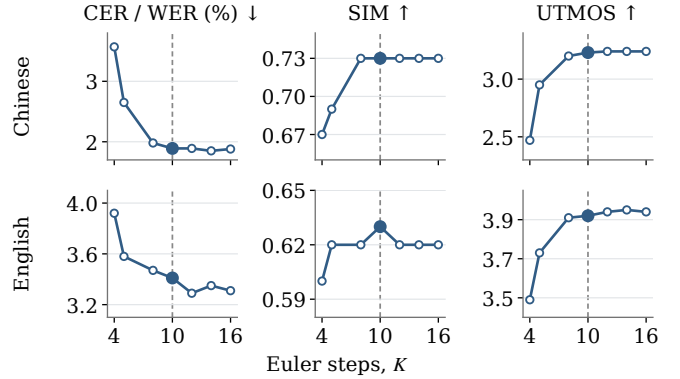


Fig. 3. Sampling-depth pilot from Joint 800k EMA with CFG=4 (two velocity branches per Euler step). The dashed line marks the selected pilot depth of $K = 10$.

preservation from overall perceived quality. Cross-system results depend on generation interfaces; matched continuations isolate supervision effects.

Learning trajectories and sampling depth. Figure 2 shows faster error reduction with joint learning, especially for Chinese. The larger Chinese gains may reflect greater room for improvement under generation-only training. Auxiliary transcription adds a content-learning signal, whose benefit may be smaller where generation-only performance already approaches that of joint training. More similar relative gains after joint pretraining (Table 1B) are consistent with this account. At 500k, joint learning lowers errors but slightly reduces English UTMOS relative to generation-only training; content accuracy and predicted naturalness need not improve together.

In the pilot (Fig. 3), increasing sampling depth from 4 to 10 steps reduces CER/WER from 3.57%/3.92% to 1.89%/3.41%; SIM and UTMOS stabilize around 8–10 steps. Beyond 10 steps, changes are small and metric-dependent. The diminishing returns may reflect errors in the learned velocity field that additional integration steps alone do not resolve. The pilot neither determines training reconstruction depth nor replaces the main 48/64-step evaluation.

5. CONCLUSION

We studied how training on speech understanding can improve speech generation. Matched bilingual experiments show that retaining shared text prediction improves content accuracy, while Reconstruct CE further improves accuracy, predicted naturalness, and speaker similarity through a frozen understanding branch. The approach connects local velocity learning with supervision of reconstructed content, without waveform decoding in the reconstruction branch or changes to inference. These results support two complementary uses of understanding supervision: shaping shared parameters through text prediction and guiding generation through feedback on reconstructed content.

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